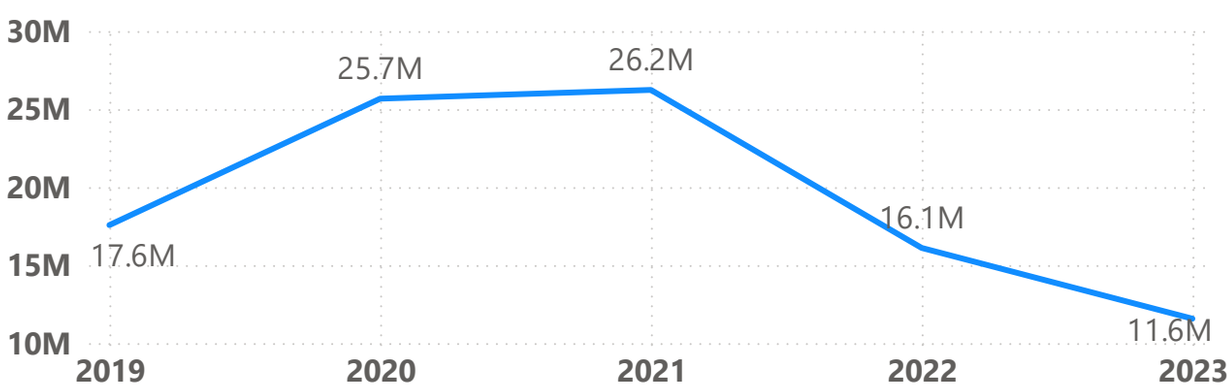
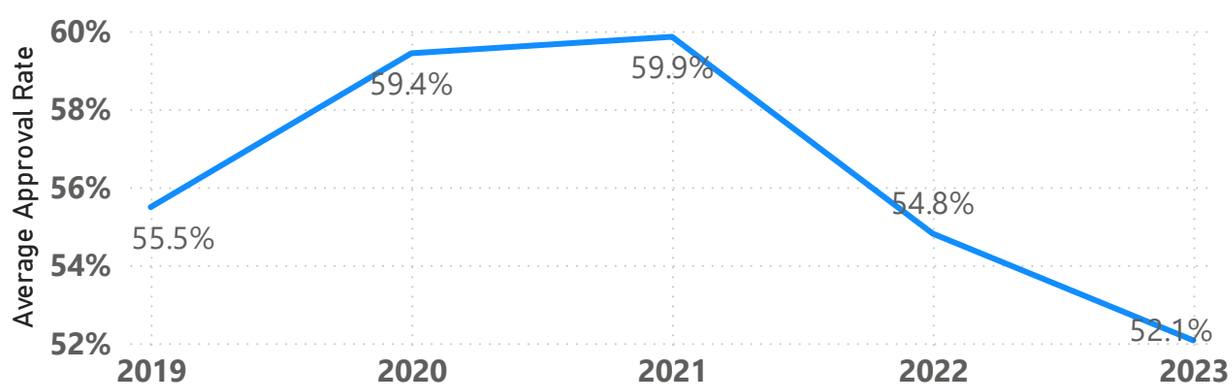
US MORTGAGE PORTFOLIO OVERVIEW (YEAR 2019 to 2023)

97M	57.2%	52.07%	2021	59.85%
Total Applications	Average Approval Rate	Approval Rate 2023	Peak Approval Year	Peak Approval Rate

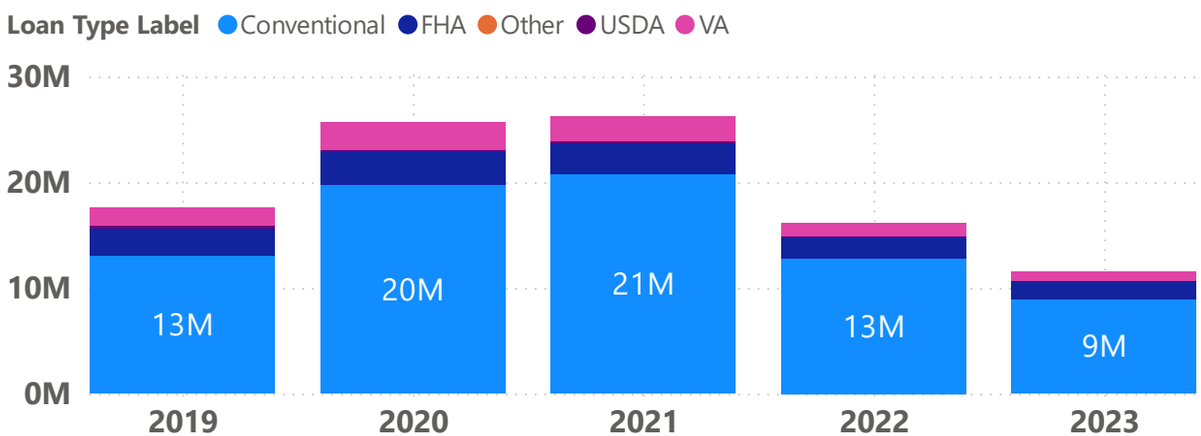
Total Applications by year



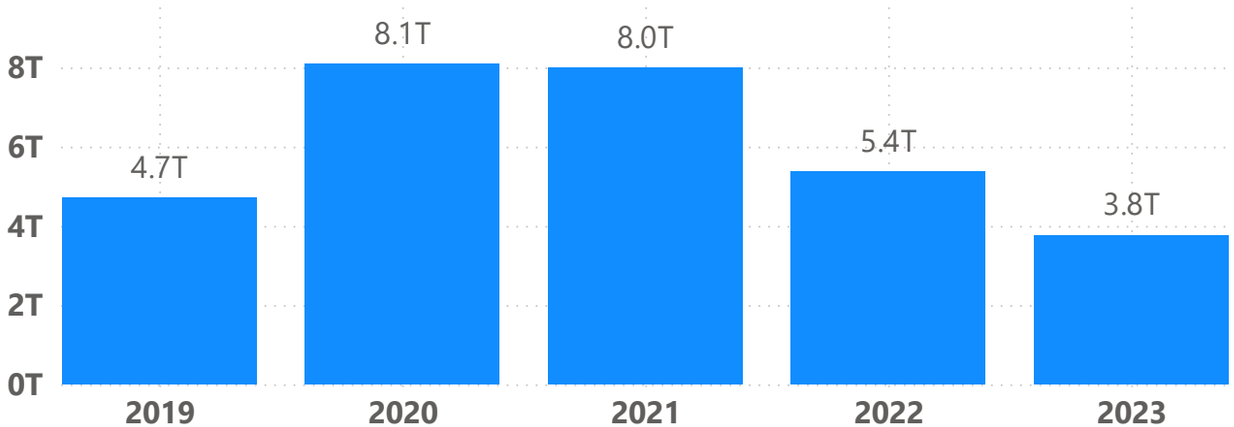
Average Approval Rate by year



Total Applications by year and Loan Type



Total Loan Volume (Trillions USD) by year



Market Expansion Followed by Sharp Contraction

Application volumes and approval rates peaked in 2020–2021 amid record liquidity, before tightening significantly in 2022–2023 as rates rose and underwriting conditions strengthened.

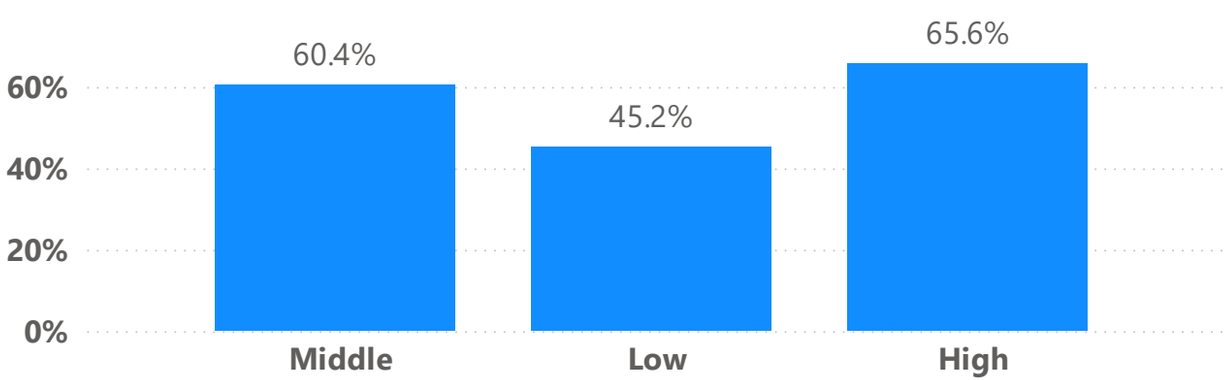
BORROWER RISK PROFILE

45.19%
Low Income Approval

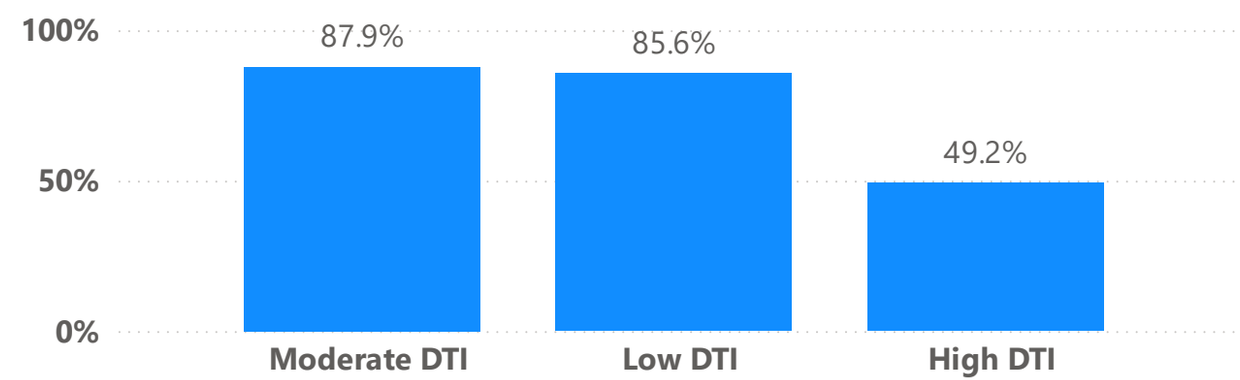
45.53%
High DTI Approval

3.01
Average LTI

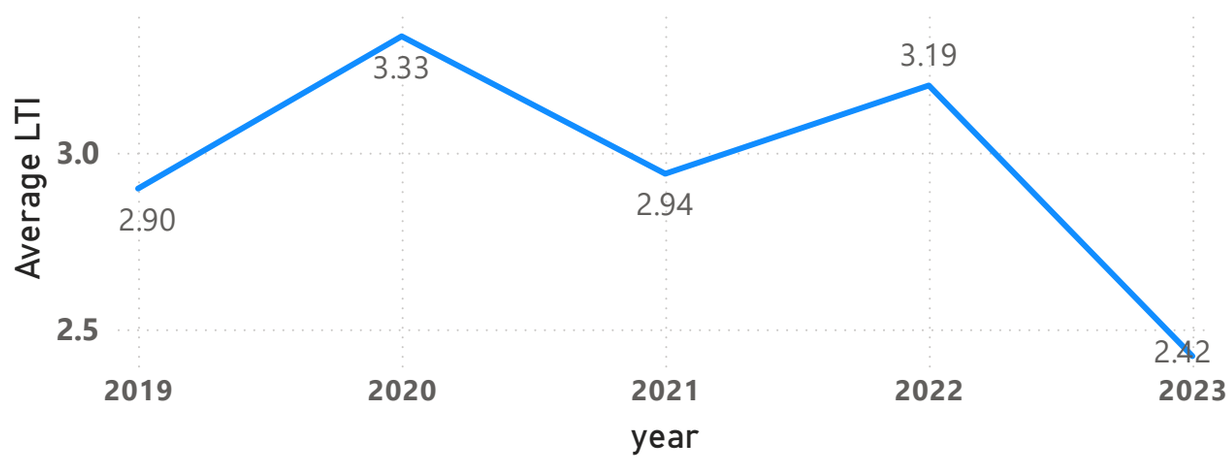
Average Approval Rate by Income Band



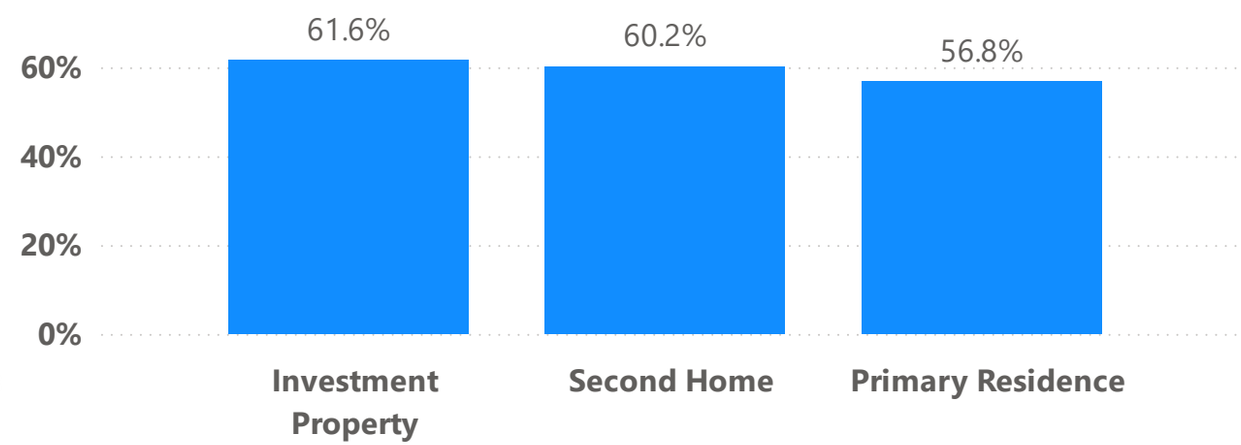
Average Approval Rate by Debt-to-Income



Average LTI by year



Average Approval Rate by Occupancy Type



Underwriting Sensitivity Concentrated in DTI & Income Segments

Approval rates decline materially for high DTI and lower-income borrowers, confirming debt burden and repayment capacity as primary credit risk drivers.

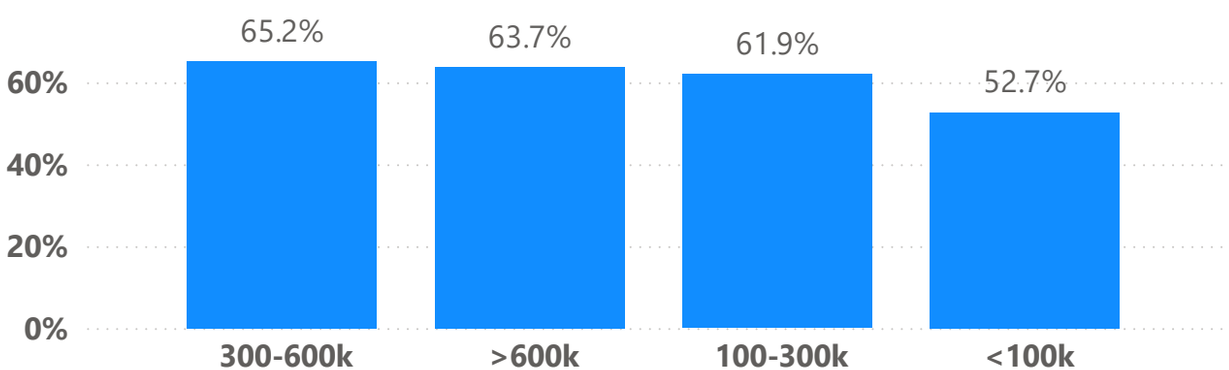
LOAN STRUCTURE & CREDIT DESIGN

308K
Average Loan Size

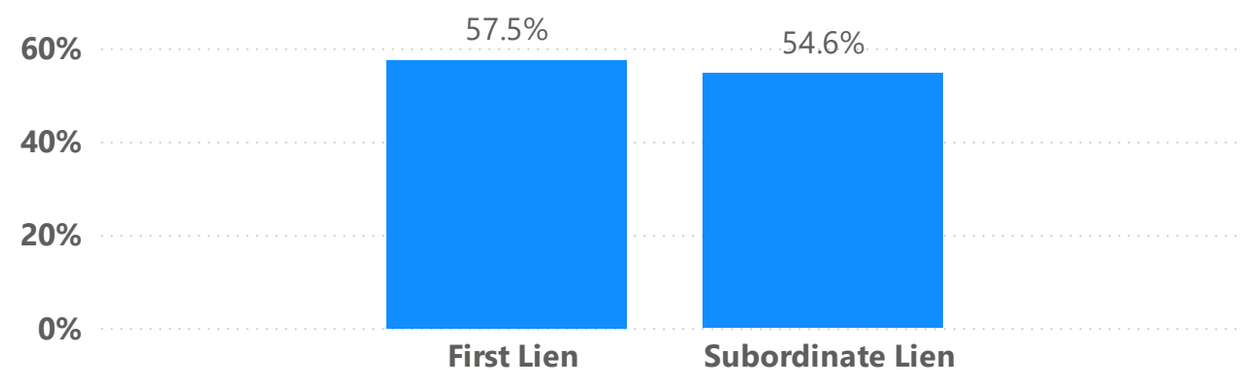
57.03%
Conforming Approval Rate

60.61%
Non-Conforming Approval Rate

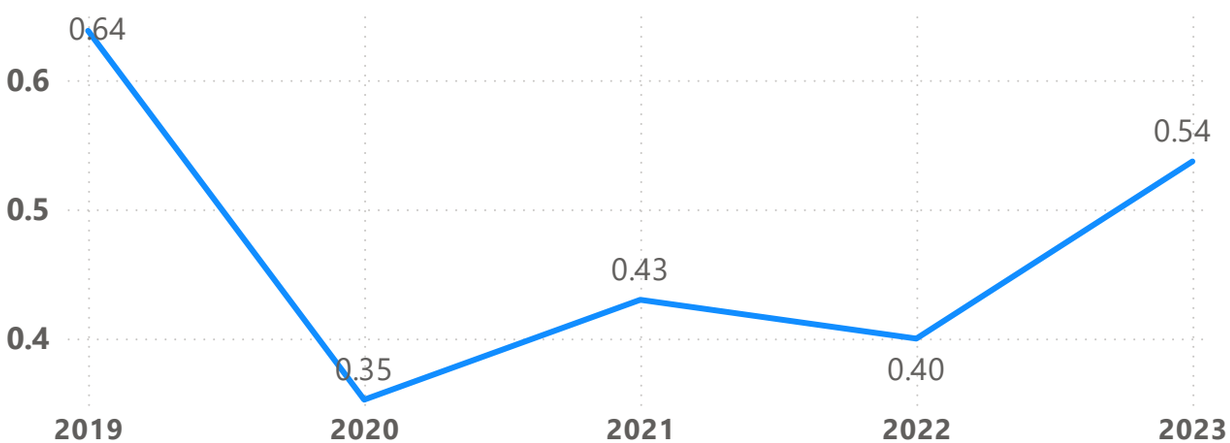
Average Approval Rate by Loan Size



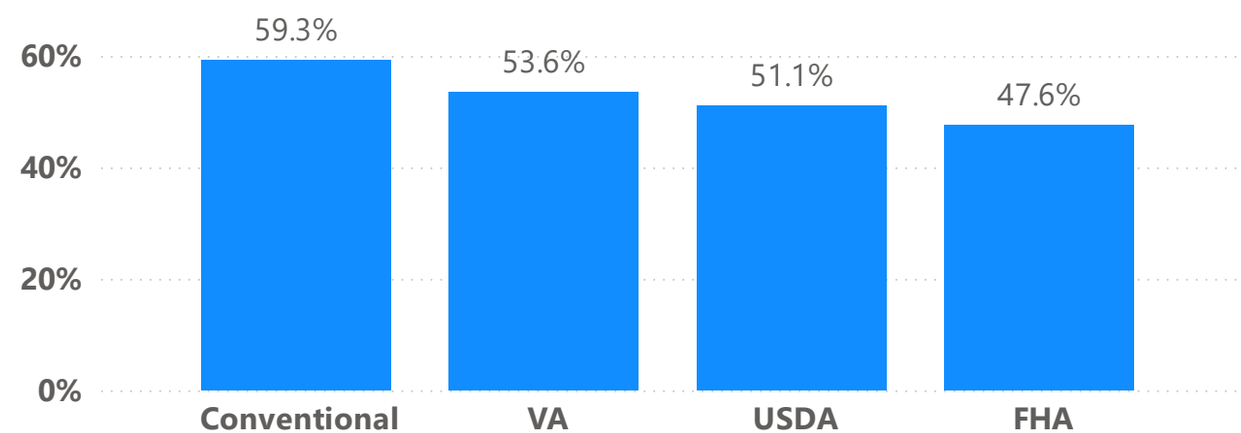
Average Approval Rate by Lien Status



Rate Spread Trend



Loan Type vs Approval



Structural Risk & Pricing Reflect Cycle Tightening

First-lien and larger loan balances maintain stronger approval rates, while rate spreads widened post-**2021 (COVID)**, indicating pricing adjustment and increased credit discipline.

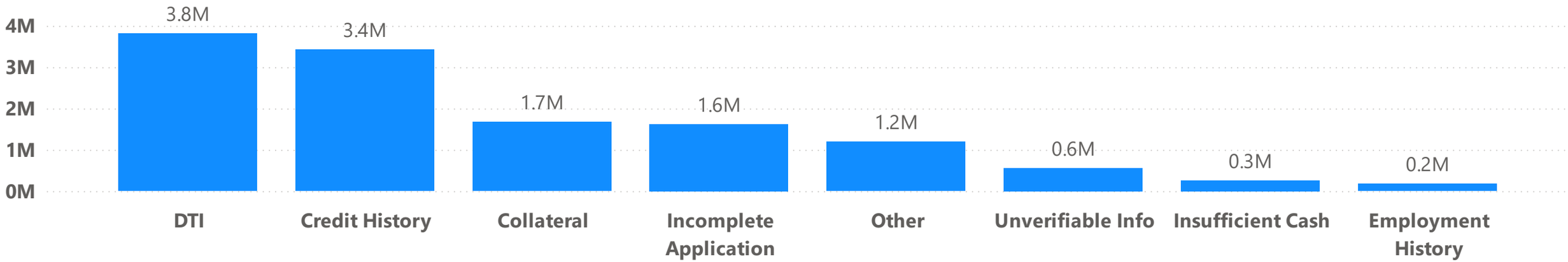
LOAN DENIAL REASONS

13.3%
Denial Rate

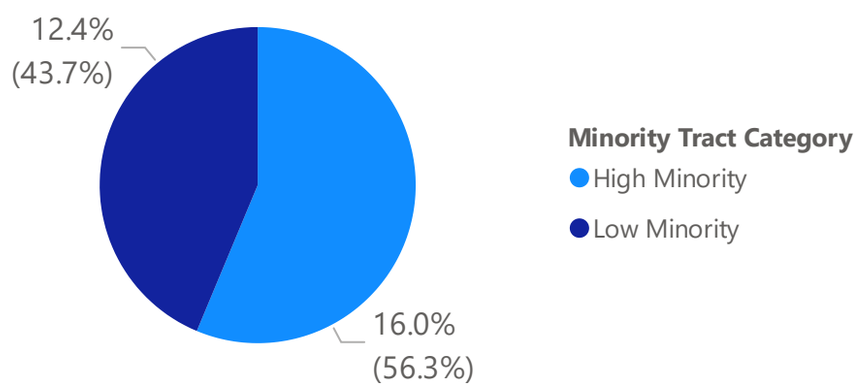
17.6%
Denial Rate 2023

25.3%
Peak Denial Rate

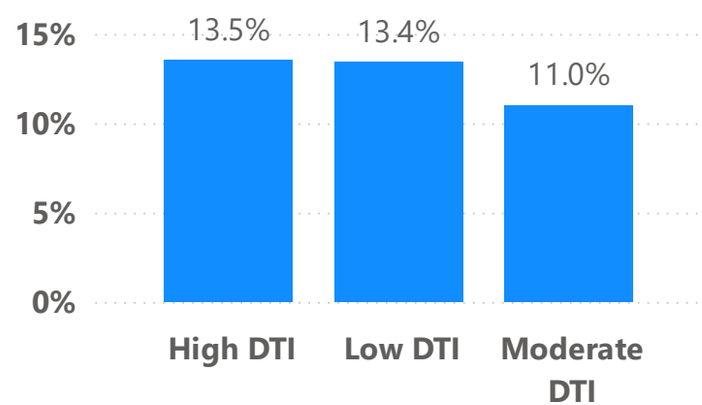
Top Denial Reasons



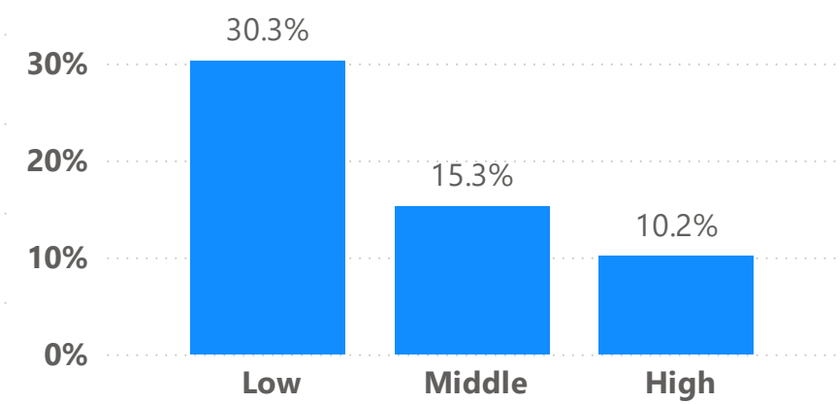
Denial Rate by Minority Category



Denial Rate by DTI Band



Denial Rate by Income Band



Denials Driven Primarily by DTI & Credit History

Debt-to-income ratio and credit history represent the dominant denial drivers, with denial rates rising into 2022–2023 alongside broader credit tightening.

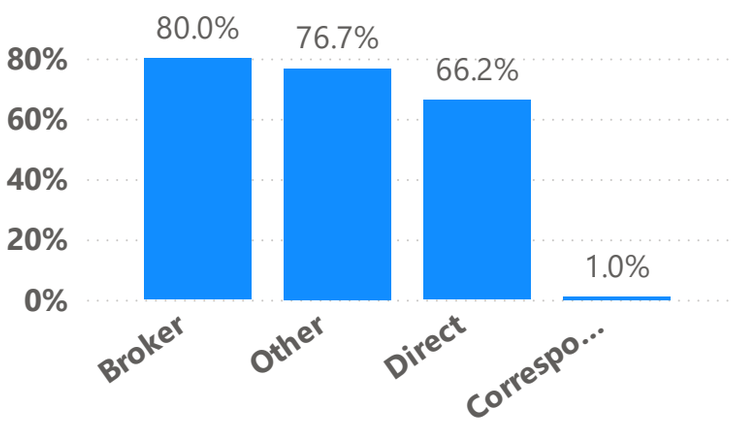
LOAN CHANNEL & SECONDARY MARKET

79.5%
Direct Origination Share

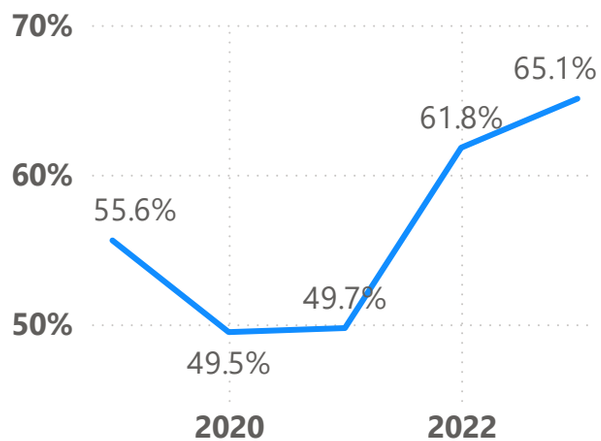
3.57%
Broker Origination Share

54.6%
Portfolio Retention Share

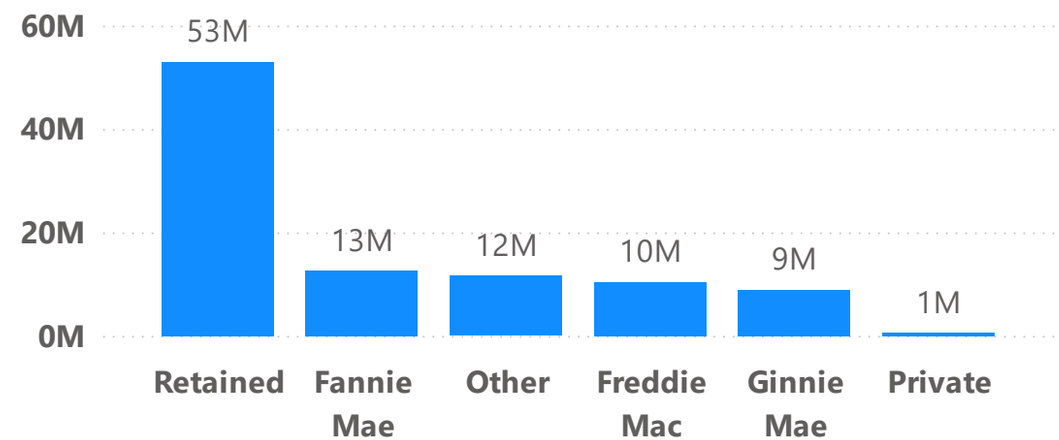
Approval Rate by Channel



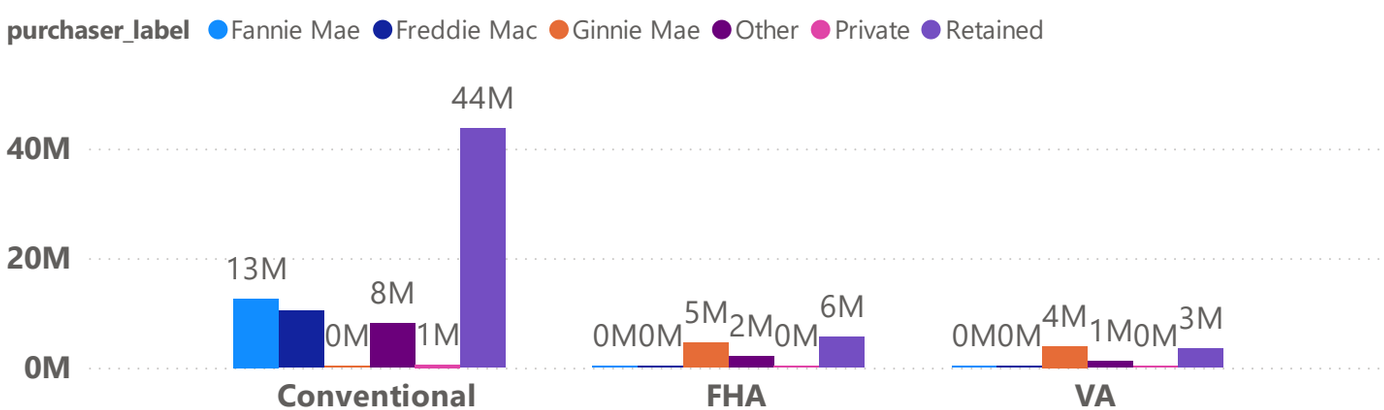
Portfolio Retention Share Trend



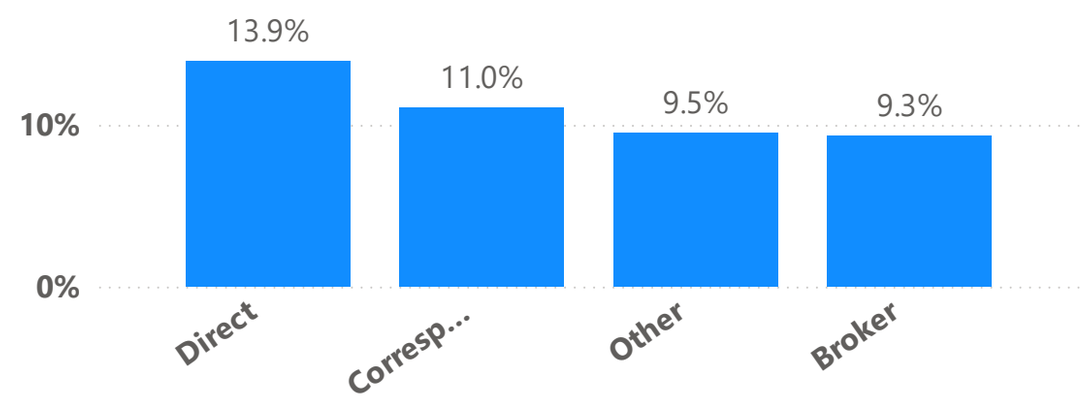
Purchaser Type Distribution



Loan Type vs Purchaser



Channel vs Denial Rate



Secondary Market Liquidity Shift Toward Portfolio Retention

Portfolio retention increased materially in 2022–2023, while conventional loans remained GSE-driven and FHA production primarily flowed through Ginnie Mae, reflecting structural capital market dynamics.

U.S. Mortgage Credit Cycle Summary

1. Market Cycle Dynamics

- 97M total applications processed
- Approval rates peaked at 59.85% in **2021** before falling to 52.07% in 2023
- Application volumes contracted sharply post-2021

The data reflects a **pandemic-era liquidity surge** followed by rapid tightening amid rising interest rates and stricter credit conditions.

2. Core Risk Drivers

- **High DTI** remains the **strongest determinant** of **denial**
- Credit history represents the second-largest constraint
- **Lower-income borrowers exhibit materially higher denial rates**
- Loan-to-income ratios peaked during the liquidity expansion phase

Underwriting outcomes remain strongly aligned with borrower leverage and repayment capacity.

3. Structural & Pricing Adjustments

- First-lien structures maintain stronger approval outcomes
- Larger balances show slightly stronger performance characteristics
- Rate spreads widened in 2022–2023

Credit discipline increased alongside macro tightening.

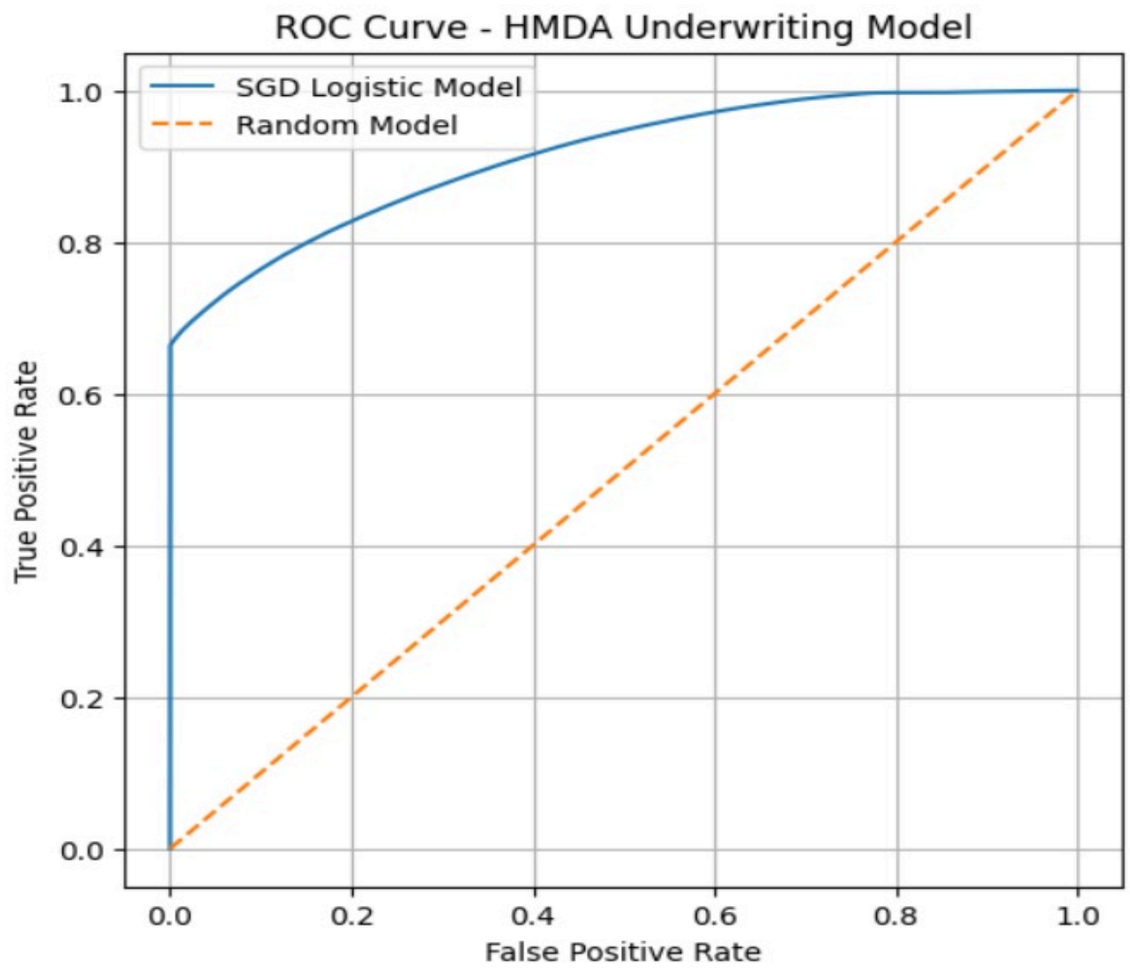
4. Secondary Market & Liquidity Shift

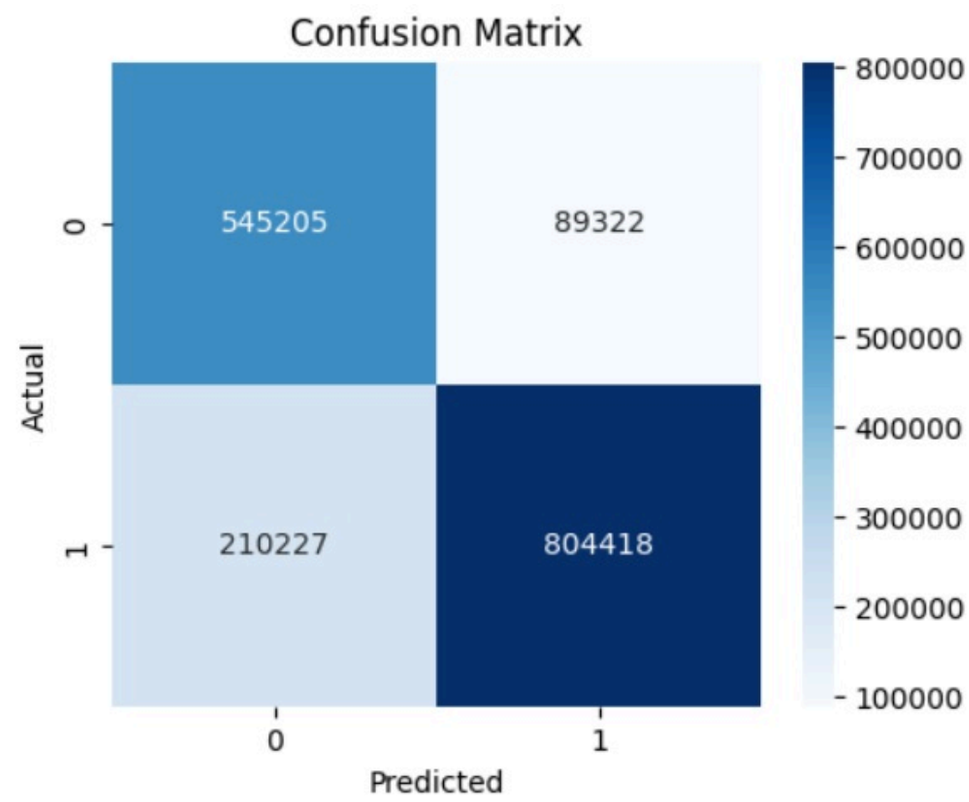
- Portfolio retention rose from ~49% (2020) to ~65% (2023)
- **Conventional** production primarily sold to **Fannie Mae & Freddie Mac**
- **FHA** production largely securitized via **Ginnie Mae**
- **Direct channel** exhibits marginally **higher denial** intensity

Capital allocation patterns reflect a measurable contraction in secondary liquidity and **increased on-balance-sheet** retention.

LOGISTIC REGRESSION MODEL PREDICTION OF LOAN APPROVAL & DENIAL

0.93	0.82	0.90	0.79	0.87
Latest ROC	**Accuracy**	**Precision**	**Approval Recall**	**Denial Recall**





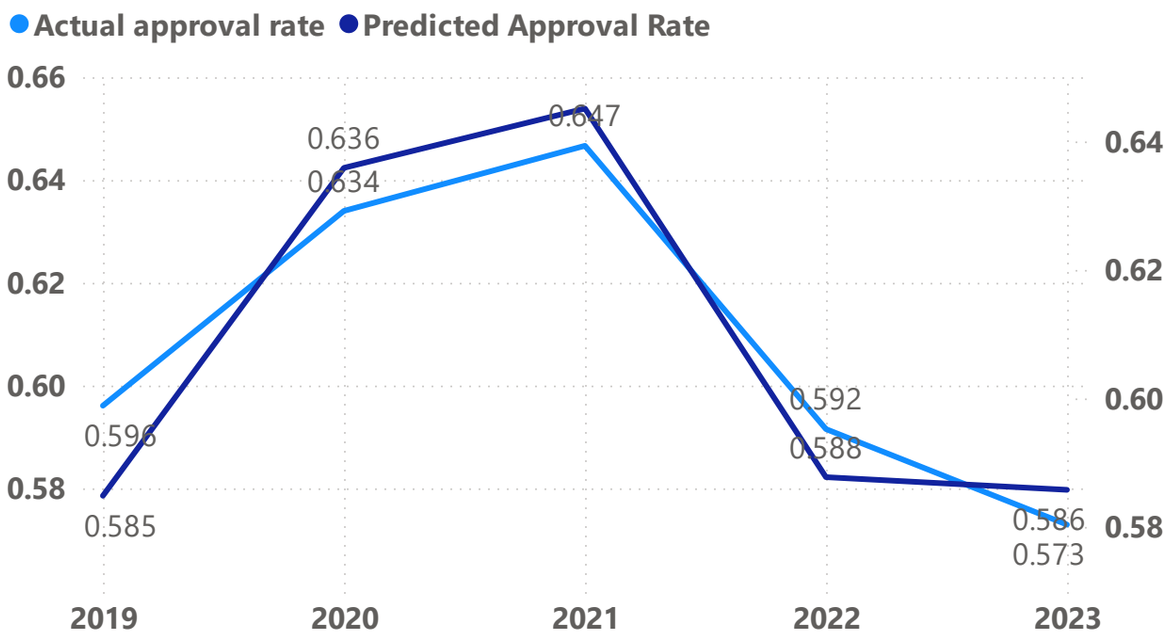
Logistic Regression (SGD optimization) trained on **8.2M mortgage applications** using **structured underwriting features**.

The logistic regression model demonstrates **strong predictive power (ROC-AUC 0.93)**, effectively separating **approved vs denied** mortgage applications.

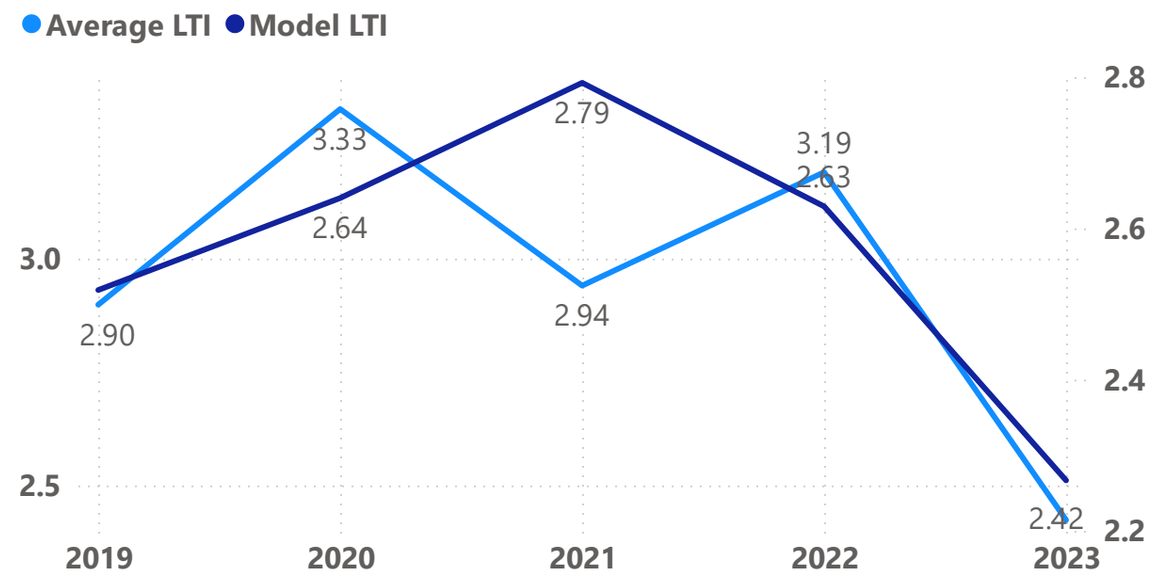
Precision above **90%** indicates that when the model predicts **loan approval**, it is **correct the vast majority of the time**.

ML MODEL PERFORMANCE TRENDS

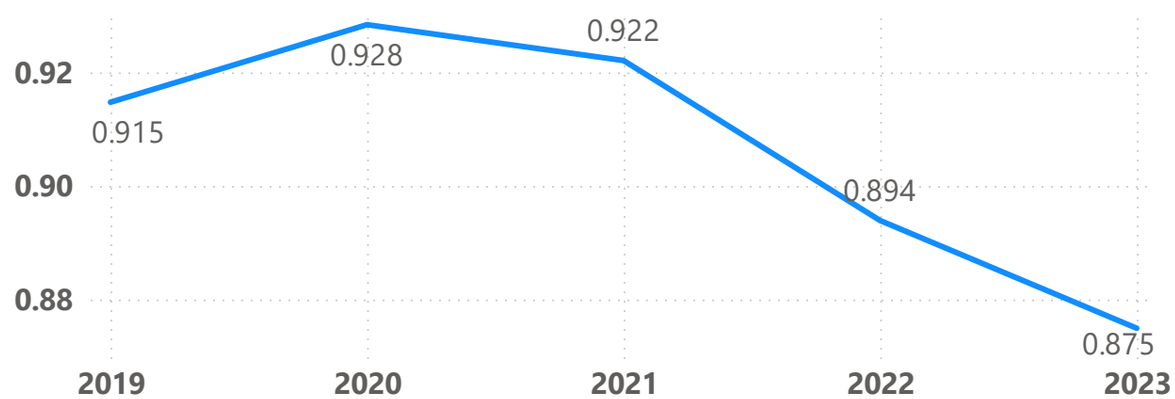
Actual Approval Rate vs Model Predicted Approval Rate



Actual Average LTI by Year vs Model Average LTI by Year



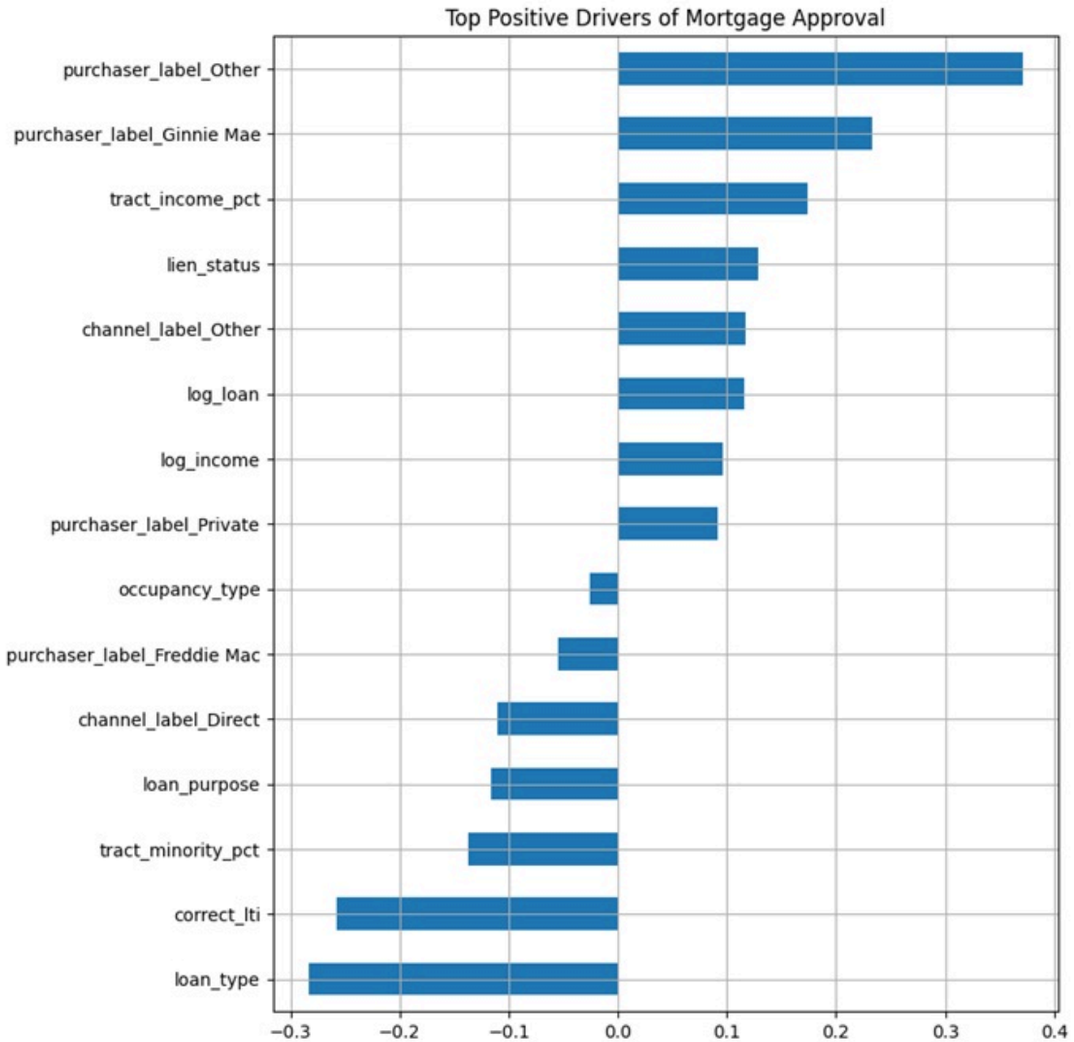
ROC-AUC by Year

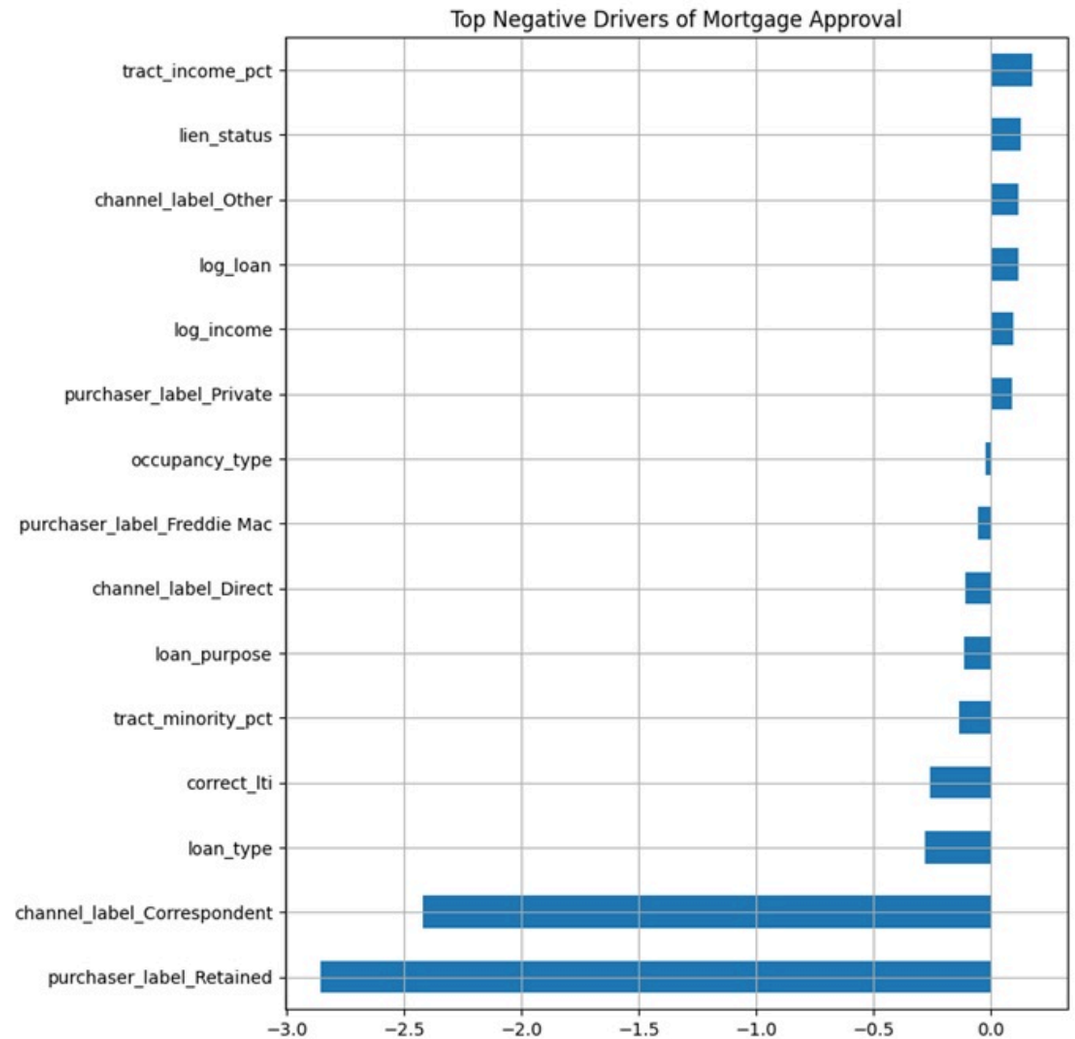


Model predictions closely track **actual approval trends across years**, indicating **stable model performance over time**.

Changes in **average loan-to-income ratios (LTI)** align with shifts in **approval rates**, reflecting the model's **sensitivity to borrower leverage conditions**.

DRIVERS OF MORTGAGE APPROVAL AND DENIAL





Borrower leverage (Loan-to-Income) and **loan structure variables** are key drivers of approval outcomes.

Higher borrower income and **government-backed purchaser channels** increase **approval probability.**

Higher leverage and **correspondent channel loans** are associated with **higher denial likelihood**.

ML Model Summary

Model Summary

The underwriting model demonstrates **strong discriminatory power** (ROC-AUC **0.91**), effectively distinguishing approved and denied mortgage applications at scale.

Business Interpretation

- **Higher** borrower **Income** materially **increases** **approval** probability
- **Higher Loan-to-Income** ratios **reduce** approval likelihood
- Government-backed purchaser channels show stronger approval patterns
- **Correspondent** channel loans exhibit relatively **lower** approval probability

Performance Snapshot

- **Accuracy**: 81.9%
- **Approval Precision**: 90.3%
- **Denial Recall**: 86.5%

Monitoring

Model Performance remains **stable** across years with no material degradation in ROC-AUC.

Business Implication

The model captures **borrower leverage** **sensitivity** and **structural channel** effects consistent with real-world
mortgage underwriting behavior.